

CO5-AT2-Q2:

Name: D.Uday Kiran

Reg no: 192425161

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Edge Coloring Problem

Problem Understanding

Edge coloring assigns different colors to connected edges of a graph. Adjacent edges cannot have the same color.

The objective is to color all edges while satisfying graph constraints.

Algorithm Used

Backtracking

Constraints

- Adjacent edges must have different colors.
- Number of colors should be sufficient.
- Invalid assignments must be removed.

Solution Approach

Steps:

Select an uncolored edge.

Try available colors.

Check whether color conflicts with adjacent edges.

Assign valid color.

Backtrack if conflict occurs.

Pruning Technique

The algorithm rejects invalid colors immediately.

Benefits:

- Reduces unnecessary search.
- Improves execution efficiency.

Correctness

The algorithm checks all possible color assignments and returns only valid edge coloring solutions.

Complexity Analysis

Worst Case:

$O(C^E)$

Space Complexity:

$O(E)$

Conclusion

Backtracking efficiently solves edge coloring by exploring possible assignments and eliminating invalid solutions early.